First error:

```
javierub@DESKTOP-CHBPSS9$ mpicc -Wall -o P3_ring P3_ring.c
P3_ring.c: In function ‘main’:
P3_ring.c:32:19: error: ‘MPI_COMM_ALL’ undeclared (first use in this function); did you mean ‘MPI_COMM_NULL’?
   32 |     MPI_Comm_rank(MPI_COMM_ALL, &rank);
      |                   ^~~~~~~~~~~~
      |                   MPI_COMM_NULL
P3_ring.c:32:19: note: each undeclared identifier is reported only once for each function it appears in
```

Second error:

```
javierub@DESKTOP-CHBPSS9$ mpicc -Wall -o P3_ring P3_ring.c
javierub@DESKTOP-CHBPSS9$ mpirun -np 2 ./P3_ring
type a word and press enter: Hallo
process 0 sends the word Hallo
process 1 has the word Hallo
[DESKTOP-CHBPSS9:07525] *** An error occurred in MPI_Send
[DESKTOP-CHBPSS9:07525] *** reported by process [1603272705,1]
[DESKTOP-CHBPSS9:07525] *** on communicator MPI_COMM_WORLD
[DESKTOP-CHBPSS9:07525] *** MPI_ERR_RANK: invalid rank
[DESKTOP-CHBPSS9:07525] *** MPI_ERRORS_ARE_FATAL (processes in this communicator will now abort,
[DESKTOP-CHBPSS9:07525] ***    and potentially your MPI job)
```

Third error:

```
javierub@DESKTOP-CHBPSS9$ mpirun -np 3 ./P3_ring
type a word and press enter: Hallo
process 0 sends the word Hallo
process 1 has the word Hallo
^C^Cjavierub@DESKTOP-CHBPSS9$ ^C
```

Javier Urbaneja Benítez 17/10/2026

- STAGE 1

① Invalid field at MPI_Comm_rank an MPI_Comm_size:
The program does not compiled at first because the first param at "MPI_Comm_rank" and "MPI_Comm_size" was "MPI_COMM_ALL" instead of "MPI_COMM_WORLD". I noticed this seeing the error when trying to compile and, when I changed it, I measured that it worked by seeing that there was not any error.

② Invalid "next" param calculation:
The program does not calculated "next" field correctly and it doesn't work. I noticed this by seeing the error "invalid rank" in the terminal output when trying to execute the program with 2 processes. I changed "next" fiel to "(rank + 1) % size" and I measured that my it works by seeing that the word pass correctly through the 2 processes.



③ wrong MC1_Rev call:
The program doesn't worked correctly when I executed it with more than 2 processes because the processes 1, 2, 3, ... always pretend to receibe the word from the process with rank 0 instead of the prev process. I noticed it when I tried to excute the program with more than 2 processes and it keeped at an infinite loop. I fixed "MC1_Rev" call by changing "0" to "prev" to use the prev process and I measured that now it works correctly because now the word pass throught all the processes correctly.